

Project 4: Ethical Considerations in Deep Learning

Name: Athul k Koshy

Reg No: 2411021240048

Description

Students will explore ethical challenges associated with Deep Learning systems, including bias, privacy, fairness, transparency, and responsible use of AI.

Task 1: Three Ethical Issues in Deep Learning

1. Bias and Discrimination

Deep learning models learn patterns from the data they are trained on, so if that data reflects historical or societal biases, the model reproduces and can even amplify them. This shows up in facial recognition systems that perform worse on darker-skinned faces, or hiring algorithms that favour certain demographics because past hiring data was itself skewed.

2. Privacy Violations

Deep learning systems, especially those trained on personal data such as images, medical records, or browsing behaviour, can memorize sensitive information. Models may inadvertently leak private data through their outputs, or be used for invasive purposes such as mass surveillance and unauthorized facial recognition.

3. Lack of Transparency (Black-Box Problem)

Deep neural networks are often so complex that even their own developers cannot fully explain why a specific decision was made. This is a serious concern in high-stakes domains like healthcare, credit lending, and criminal justice, where affected individuals deserve to understand and contest decisions made about them.

Other ethical issues that could also be considered include accountability gaps, misuse of AI for deepfakes and misinformation, job displacement due to automation, and the environmental cost of training very large models.

Task 2: Real-World Example

Example: COMPAS Recidivism Prediction Algorithm (used in U.S. courts)

COMPAS was a machine learning tool used by courts in the United States to predict the likelihood that a defendant would reoffend, and its risk scores influenced bail and sentencing decisions. An investigation carried out by ProPublica in 2016 found that the algorithm was almost twice as likely to falsely label Black defendants as high-risk compared to white defendants, while white defendants who did reoffend were more often mislabeled as low-risk.

This case raised several important ethical concerns:

- **Encoded bias:** The model reflected racial bias already present in historical arrest and sentencing data.
- **Lack of transparency:** Defendants had no way to view or challenge how their risk score was calculated.
- **High-stakes impact:** A statistical tool directly influenced decisions about a person's liberty without adequate human oversight or a clear appeal mechanism.

The COMPAS case became a landmark example of how deep learning and machine learning systems can cause real, disproportionate harm when deployed without careful auditing and accountability.

Task 3: Measures to Address the Issue

- **Diverse and representative training data:** Audit datasets for demographic imbalance before training, and actively collect data that represents all groups fairly.
- **Bias testing and fairness metrics:** Regularly evaluate models using fairness metrics such as demographic parity and equalized odds, both before and after deployment.
- **Explainable AI (XAI) techniques:** Use interpretability tools such as SHAP or LIME so that decisions can be explained to affected individuals, especially in high-stakes contexts.
- **Human-in-the-loop oversight:** Ensure that AI-generated decisions in sensitive areas such as justice, hiring, and lending are reviewed by humans rather than applied automatically.
- **Regulatory and ethical review boards:** Establish independent audits and ensure compliance with regulations such as the GDPR and the EU AI Act before deployment.
- **Transparency and accountability:** Organizations should disclose when AI is used in decision-making and provide clear mechanisms for individuals to contest outcomes.

Conclusion

Deep learning systems offer significant benefits, but they also introduce serious ethical risks around bias, privacy, and transparency. The COMPAS case illustrates how these risks can translate into real harm when systems are deployed without sufficient oversight. Addressing these challenges requires a combination of technical measures, such as fairness testing and explainability, along with organizational accountability and appropriate regulation, so that deep learning systems are developed and deployed responsibly.